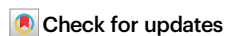


Mapping cell dynamics in human ex vivo hair follicles suggests pulling mechanism of hair growth

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Homeostasis in continuously renewing organs like the gut, skin, and hair follicles relies on niche architecture and intricately orchestrated cellular dynamics, encompassing proliferation and migration. Here, we develop a 3D live imaging system to map single cell dynamics in micro-dissected human hair follicle segments during anagen phase in ex vivo culture. This approach reveals a spiral-like downward movement of outer root sheath cells entering the lower bulb region. We further mapped upwards flowing cells in the bulb, whose pathways align with differentiation patterns observed for stem cell-derived progenitors in rodent hair follicles. Cell division rates and orientation in the bulb correlate with the velocities of the adjacent upward-moving cell layers, with faster outer layer movement linked to higher mitotic rates. Integrating fluid dynamics simulations and experimental manipulation of mitosis and actin kinetics, we propose a mechanistic model, where a pulling force induced by the outer root sheath contributes to hair fiber extrusion.

Tissue morphogenesis comprises complex cellular dynamics depending on signalling and physical cell parameters such as growth, mechanical properties and applied forces^{1,2}. Live imaging on 3D tissues, together with progresses in numerical modelling led to a better understanding of morphogenetic processes such as epithelial folding or appendage and limb bud formation^{3,4}. The hair follicle is a commonly studied mini organ used to understand tissue dynamics, morphogenesis, homeostasis and stem cell differentiation^{5,6}. However, our knowledge of hair follicle biology is predominantly based on rodent's hair follicles, while observations on the human hair follicle, exhibiting distinct characteristics, are scarcer^{7,8}.

The process of mammalian hair cyclic regeneration distinguishes three phases, the growing phase (anagen) characterized by high mitotic activity in the hair matrix and growth of the hair shaft, a regression phase (catagen) and a resting phase (telogen)^{9,10}. The architecture of the anagen hair follicle consists of seven concentric internal layers and an outer root sheath (ORS) (Fig. 1a, b)^{11,12}. These layers have different protein expression patterns and can be visualized

using immunolabeling, allowing to quantify the follicular architecture via measuring the characteristic layer diameters (Supplementary Fig. 1a, b). Lineage tracing on rodents allowed mapping the matrix progenitor cells located in a germinative layer juxtaposed to the dermal papilla, which are organized in a radial configuration with the most distal cells contributing to the most inner layers of the hair shaft and the most proximal progenitors contributing to the outer layers^{13–17} (see white arrows in Fig. 1c). Further, the ORS cellular homeostasis is controlled by a regional proliferative mode of growth on the ORS suprabulbar region^{13,14,18}.

Recent advances in live imaging allowed to trace locally the migration and cell division of cells within the mouse follicle in vivo, corroborating the compartmentalization of stem-cell niches in the bulge^{18–21}. While human follicles share similarities with those of rodents, they also show differences particularly in cycling and hormonal regulation⁷. In general, human hair follicle dynamics are less studied due to ethical and technical limitations^{7,8}. One approach for investigating human hair follicles is by maintaining them ex vivo, after

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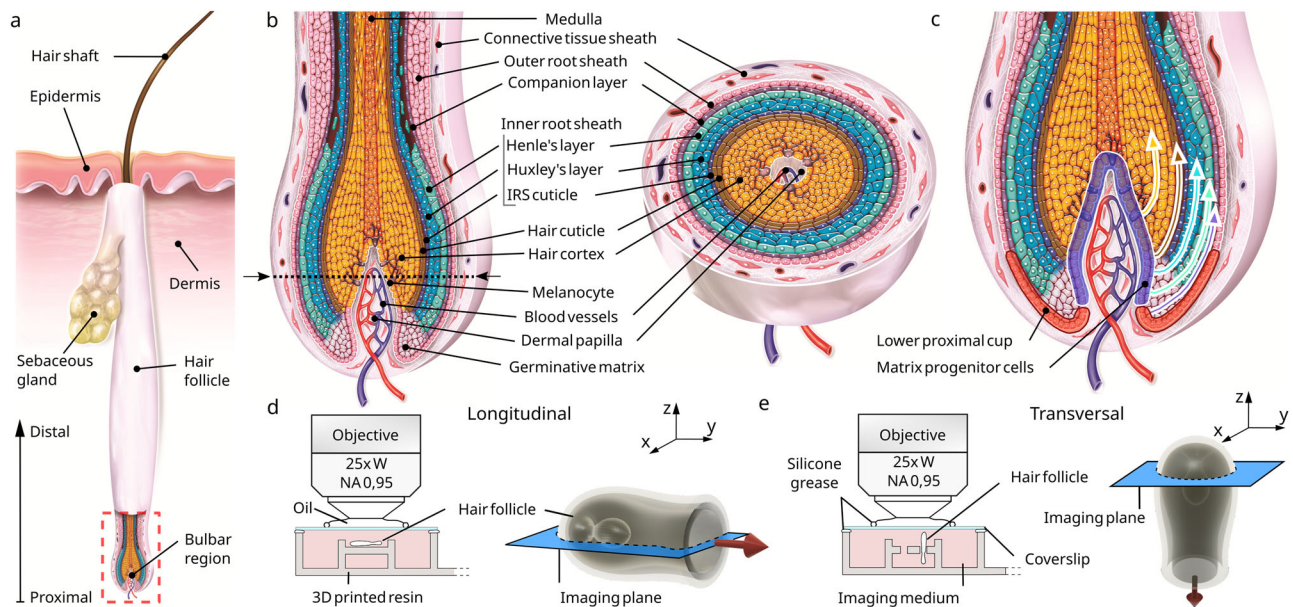


Fig. 1 | The hair follicle organization and the multiphoton setup for live-imaging of human hair follicles. **a** Scheme of human skin containing a single hair follicle. **b** Zoom of red dotted area from (a) of the hair follicle bulb containing concentric layers of different cell types. The layers are organized into different compartments: The hair shaft, which is composed of medulla, cortex and cuticle; the inner root sheath (IRS), composed of IRS cuticle, Huxley layer and Henle layer;

Companion layer, and the outer root sheath (ORS). Dotted line depicts the region for the cross-section. **c** Current model based on lineage tracing linking locally different matrix progenitor cells to the growing hair follicle layers. **d, e** Custom-built sample holder, allowing the hair follicle to be imaged horizontally leading to a longitudinal imaging plane (**d**) or vertically providing a transversal imaging plane (**e**). NA, numerical aperture; W, water immersion objective.

microdissection of tissues from cosmetic surgery^{22,23}. This approach has some limitations but also different advantages as compared to animal-based models (See Supplementary Discussion). Importantly, ex vivo hair follicles transition towards a catagen or pseudo-catagen state after a few days in culture and can therefore not reproduce the hair cycle^{7,22}, likely due to the microdissection that removes parts of the hair follicle microenvironment²⁴. Despite these limitations, the ex vivo human hair follicle culture approach has been key to study keratin synthesis^{25,26}, chronobiology^{27,28}, cell metabolism^{29,30}, cytokine biology^{31,32} and melanin pigmentation^{33–36}. It also allowed for insights in the causal factors determining hair shape³⁷ and mechanics³⁸ and it has led to a better understanding of the impact of signaling molecules and other regulatory factors on hair growth^{23,39} allowing its use for drug testing within the known associated limitations (Supplementary Discussion)^{40–43}.

While today local cell dynamics, especially in rodents are known^{13,14,18–21}, a global map of human hair follicle dynamics at the single cell level is missing. The goal of this study is to investigate whether the migratory patterns in a global map of human hair follicles align with the current model of spatially distinct progenitor cells that are feeding different hair follicle layers (Fig. 1c). In addition, the quantification of global cell dynamics allows to determine the main drivers for hair follicle growth using physical models. To reach this, we use live imaging in three dimensions (3D+t) on ex vivo human anagen hair follicle segments with single cell tracking and particle image velocimetry (PIV) to map the cell dynamics of a growing hair follicle for more than one day. We observe different speeds for the upward moving cortical and IRS layers and a spiral like downward movement of the ORS cells, with cells of the bulb outer layer entering the lower (proximal) bulb region. Quantification of these flows and the rate of cell-divisions in the matrix allow to build a numerical model using fluid dynamics. The model can predict the experimentally measured velocity profiles when an active force acting on the outer layers is assumed in addition to the occurring cell divisions in the matrix. Based on

experiments interfering with mitosis and actin kinetics in the follicle we propose that an actin-mediated pulling force applied by the ORS on the adjacent companion layer contributes to hair fiber growth.

Results

Hair follicle cell dynamics in the supra-bulbar region

To investigate the cell dynamics of growing human hair follicles, we developed an ex vivo imaging method where a hair follicle is placed in a custom-built microscope holder under a multi-photon microscope (Fig. 1d, e). To this end, we micro-dissected human scalp hair follicles gained from facelift surgery to obtain non-pigmented, terminal hair follicles. To visualize single cells, we acquire either the auto-fluorescence signal (AF), the second and third-harmonic generation (SHG, THG) or the exogenous fluorescence from nucleus staining (Hoechst-33258). The auto-fluorescence arises mainly from the cytoplasmic NADH, accompanied by a lesser contribution from FAD, keratin, and others⁴⁴. The follicles are placed in the microscope holder either horizontally (Fig. 1d) or vertically (Fig. 1e) in culture medium at 37 °C and 5% CO₂ and imaged in 3D every 30 min for up to 90 hours with a 4–5 μm z-step (See Methods). In our experimental conditions, we observe first structural changes in the follicle architecture reminiscent of the anagen – pseudo-catagen transition at median times of ca. 20 hours of live-imaging (Supplementary Fig. 2) and thus constrict analysis and our cell tracking measurements to times before first signs of catagen (Methods).

First, we tracked the cells in the supra-bulbar region, at different radial positions within the follicle (Fig. 2a). In this example, the cells of the anagen hair follicle positioned at the same height at $t = 0$, migrated differently during 18.5 h (Fig. 2a, first row, Supplementary Movie 1). Their movement depends on their radial position, they can migrate upwards (green and yellow cells) or downwards (red cell). During live imaging, we can define the corresponding layer in which a single cell is embedded based on the relative position of the cell (green cell located in cortex, yellow cell in IRS, red cell in ORS, Supplementary Methods,

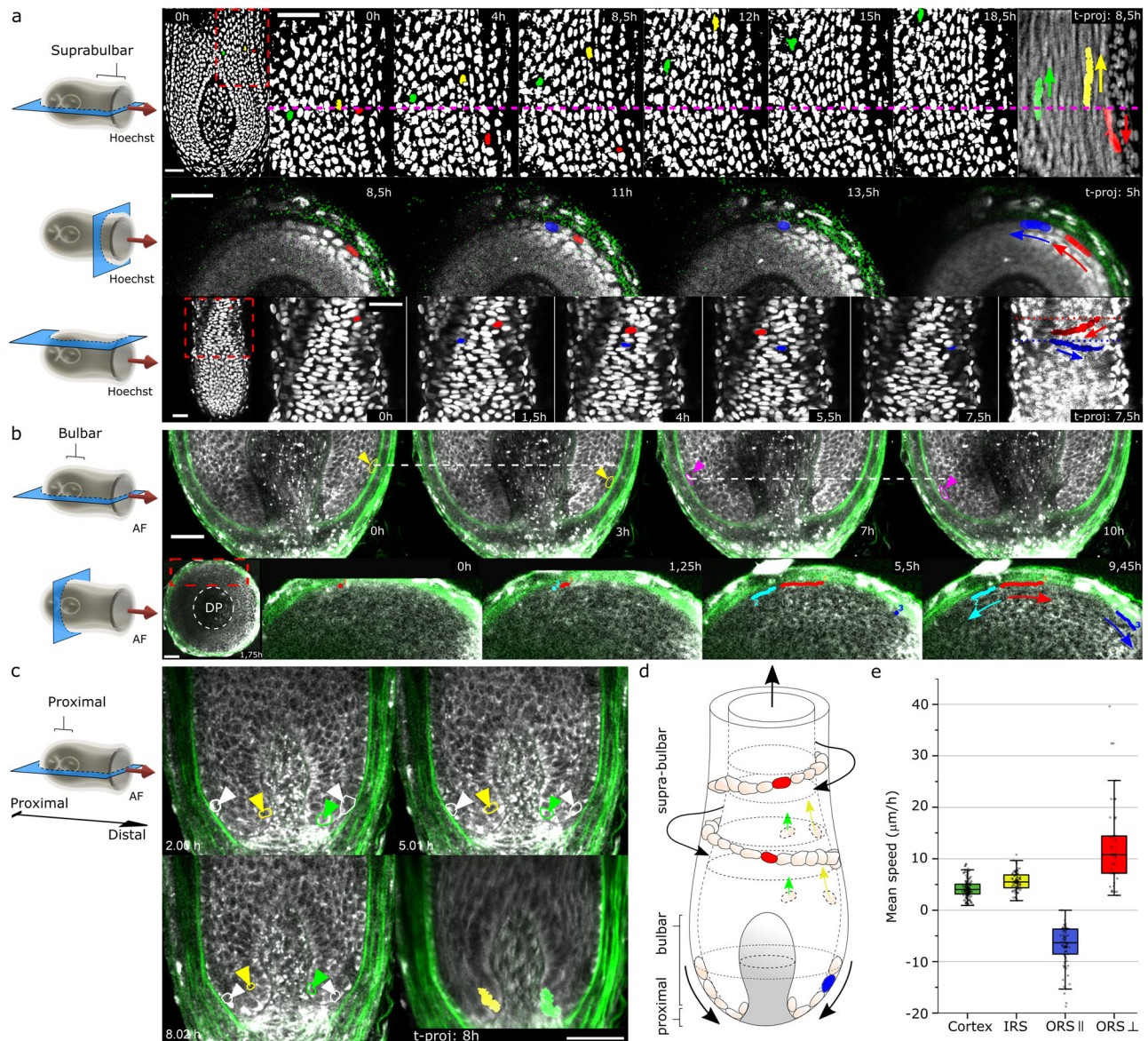


Fig. 2 | Dynamics of hair follicle cells. a Cellular dynamics on the supra bulbar region. Last image in a row is time projection. First row: Longitudinal imaging of a human hair follicle with nuclei labeled by Hoechst. Dashed purple line highlights the starting height for different cells: green – cortex cell, yellow – IRS cell, and red – ORS cell. Second row: transversal imaging of a hair follicle tracking the rotational movement of two ORS cells (red and blue). Third row: Longitudinal image, taken on the ORS layer. Blue and red cells show clockwise and counterclockwise rotational downwards movement. **b** ORS cell dynamics on the bulbar region. First row: Longitudinal image with two highlighted ORS cells moving downwards. Second row: Transversal image with three cells (dark blue, red, light blue) illustrating the rotational movement of ORS. **c** ORS cell dynamics on proximal region of the hair bulb. Longitudinal optical section where four different ORS cells are tracked: White ORS cells moved along the bulb ORS down to the lower proximal cup (LPC); yellow cell from the LPC detached the germinative layer juxtaposed to the DP; green cell initially in the LPC moved upwards along the germinative layer, keeping the contact

with the DP. **d** Model schematizing the observed movements. Supra-bulbar ORS cells move in a direction perpendicular to the proximal-distal axis (red ORS cells, ORS_⊥) or parallel to that axis, (blue cells, ORS_{||}) when around the bulb. **e** Boxplot of cell speeds in the different layers from AF and Hoechst follicles for: Cortex cell speed ($n = 144$ points, each representing the mean velocity of a single tracked cell in the cortex of a follicle. Sample size is $N = 20$ hair follicles from $D = 14$ different donors), IRS speed ($n = 91$, $N = 20$, $D = 13$), ORS parallel speed ($n = 79$, $N = 18$, $D = 13$), and ORS perpendicular speed ($n = 57$, $N = 10$, $D = 8$). Boxes denote central half of datapoints with the inner line being the median. Whiskers show furthest data point within 1.5 times the inner quartile length. All scale bars are $50 \mu\text{m}$. AF, Auto-fluorescence; C, Cortex; DP, dermal papilla; GCs, germinative layer cells. Red dashed boxes depict the zoomed area of the following images, green signal is SHG. Repeatability of all observations shown in **a–c** are detailed in Supplementary Information. Source data are provided as a Source Data file.

Supplementary Fig. 1c–d). In the supra bulbar region, the average relative speed of the IRS layer is $1.8 \pm 1.4 \mu\text{m/h}$ faster as compared to the cortex cells ($n = 156$ cell tracks, $N = 17$ follicles, $D = 11$ donors, see Methods and Supplementary Fig. 3).

The downwards moving cells are directly adjacent to the connective tissue sheath and correspond to the ORS layer. Following these

cells in a longitudinal plan is challenging, since the ORS cells often leave the imaging plane, suggesting a more complex 3D movement. To get more insight, we performed transversal imaging (Fig. 1e) and observed that some single ORS cells follow a rotational migration perpendicular to the hair follicle growth axis (Fig. 2a, middle row; Supplementary Movie 2). To further clarify this, we set the longitudinal

imaging plane specifically on the ORS layer (Fig. 2a, third row, Supplementary Movie 3), where we observed clockwise and counter-clockwise helical like downward movements. These perpendicular ORS speeds (ORS $\perp$) averaged $12.0 \pm 7.7 \mu\text{m/h}$ across at least 3 tracked cells in each of $N=10$ follicles from $D=8$ independent donors (Supplementary Fig. 3 and Supplementary Movie 4).

ORS cells enter the bulbar compartment

We next imaged the hair follicle in the bulbar region (Fig. 2b). In this region, we still observe both: rectilinear (upper row) and rotational (lower row) downwards movement of ORS cells (Supplementary Movie 5 and 6). Tracking at least 3 individual cells within the ORS of $N=18$ independent follicles obtained from $D=13$ different donors show an average downwards speed (ORS $\parallel$) of $-6.9 \pm 3.8 \mu\text{m/h}$ (See Methods and Supplementary Fig. 3). This downward migration of ORS cells in human hair follicles agrees with observations in rodents²⁰, where especially in whiskers an average downwards speed of ORS cells was estimated to be around $4 \mu\text{m/h}$ ⁴⁵. However, when imaging further down, at the lower proximal cup, we could not observe rotational ORS cell migrations while rectilinear downward movements were still registered (Fig. 2c white arrows). This indicates a transition zone between the supra-bulbar region and the lower proximal cup where the circular downward movement becomes more and more rectilinear. This agrees with a quantification of the angles between the horizontal axis and the direction of the rotational ORS cell movement: we define that a positive angle between 0° and 90° corresponds to a left or right-handed upward movement and that a negative angle between 0° and -90° corresponds to left or right-handed downward movement. In the supra bulbar region, we measure in average a rather horizontal movement with an angle of $-5^\circ \pm 19^\circ$ ($n=19$, $N=4$, $D=3$, Supplementary Movie 4) while the movement becomes steeper in the bulbar region ($-38^\circ \pm 20^\circ$, $n=16$, $N=3$, $D=3$, Supplementary Movie 7).

We quantified the cell speeds of IRS, cortex, and ORS both in parallel and perpendicular directions at the locations indicated by the colored cells in Fig. 2d. Our results show that in the region above the dermal papilla, IRS cells move faster than cortex cells (Supplementary Fig. 3l–n). ORS cells, on average, migrate downward along the proximal-distal axis showing an additional perpendicular movement. The boxplots in Fig. 2e summarize these cell speeds where the colors correspond to the color of highlighted cells in Fig. 2d. Additional details on quantification methods, definition of measurement location, replicates, and an analysis of follicle-to-follicle variability can be found in the Methods section and in Supplementary Fig. 3. These data demonstrate that in human hair follicle segments, the ORS cells show locally distinct cell migration patterns, with the supra-bulbar ORS moving almost perpendicular to the proximal-distal axis, while the cells of the bulb outer layer migrate more in parallel to the proximal-distal axis, entering the bulbar region.

Quantification of mitosis and velocity fields in the bulb

The hair follicle is one of the most proliferative tissues in the body and its growth is fueled by cell divisions in the matrix. To quantify the mitoses occurring in the progenitor cells, we imaged follicles with stained nuclei allowing the observation of the metaphase plate in the longitudinal imaging setup (Fig. 3a, b). The position, time and orientation of each cell division was tracked in a portion of the 3D z-stack, to build a mitosis density map based on the quantification of nearest neighbors (Fig. 3c). The majority of cell divisions occur in the hair matrix below the Auber line⁴⁶, consistent with previous findings in rodent and anagen human hair follicles^{13,39,47}; however, cell divisions are also observed above the Auber line in transient-amplifying cells^{13,48,49} (Fig. 3c). Division rates were higher in the proximal half of the matrix (13.1 ± 4.2 mitoses/h) as compared to the distal half (5.7 ± 2.5 mitoses/h, Fig. 3d). With 30% of the total volume quantified, we get a

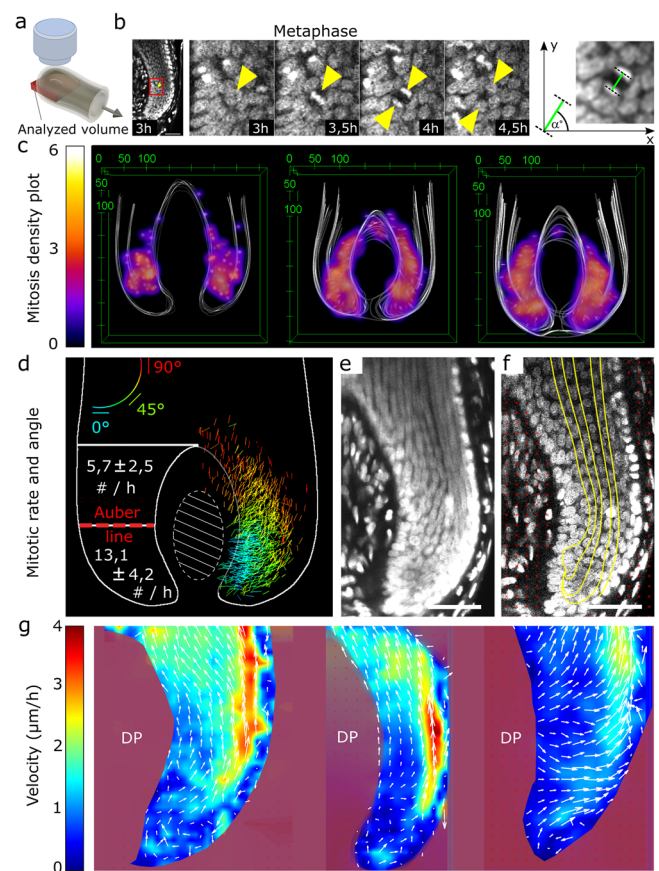


Fig. 3 | Dynamics and orientation of cell divisions in the matrix and quantification of adjacent cell flow. **a** Sketch showing the volume of the hair follicle considered for the analysis of mitosis, close to the equatorial plane. **b** Example of a matrix cell undergoing mitosis. Red square corresponds to zoom shown in the following images. During metaphase, the chromosomes stained with Hoechst form the visible metaphase plate highlighted by the yellow arrow. This allows to follow the number and orientation (α) of cell mitosis spatio-temporally. **c** 3D distribution images of the density of mitosis acquired during 14.5 h in the observed volume from three different hair follicles. The heatmap translates the number of closest mitoses in the vicinity of a pixel within a weighted radius of $10 \mu\text{m}$. **d** Diagram summarizing the angles (right half) and frequencies (left half), given in mitosis per hour (#/h) within the observed volume ($N=4$ hair follicles). Assuming axisymmetry, the angles are all reported on the same half of the follicle. Red dashed line denotes the Auber line. **e** Averaged time-projection of 10 images acquired every 30 min from the bulbar region, the resulting bright and dark streamlines are visualizing the cell flow. **f** Streamlines on the same follicle as determined via particle image velocimetry (PIV, $N=7$). **g** Velocity field of the bulbar region of three hair follicles determined via PIV analysis. Speed vectors are shown in white; the color code reflects the velocity magnitude. Scale bars, $50 \mu\text{m}$, DP, dermal papilla.

total average rate of 62.7 mitosis/h for the entire hair follicle matrix ($N=4$).

To assess the relative motions connected to the cell divisions, we plot their orientation according to the cell position in the matrix (Fig. 3d). The orientation of cell division depends on their location along the dermal papilla: while this orientation is almost perpendicular to the surface of the dermal papilla in the proximal half of the matrix, it becomes gradually parallel to the follicle growth axis in the distal half of the matrix. The averaged time projection of longitudinal images demonstrates the streamlines of upwards moving cells (Fig. 3e, Supplementary Movie 8). These streamlines can also be determined using particle image velocimetry (PIV) (Fig. 3f), which allows in addition to measure the corresponding velocity fields of the upwards moving layers (Fig. 3g). The orientation of cell divisions matches the flow direction of adjacent cells (Fig. 3d–g). We observed “onion-shaped”

velocity fields where cells feeding the IRS layer show higher migration speeds as compared to cells forming the cortex. This observation aligns with the higher mitotic activity of the proximal region of the matrix, feeding the faster moving IRS layer with more cells as compared to the cortex, and this also aligns with the higher speed of the IRS layer cells relative to the cortex cells in the supra-bulbar region (Fig. 2a). The resulting cell migration streamlines are consistent with earlier *in vivo* findings in human hair follicles⁴⁹ and further support the current model of differentiation and migration paths of the IRS and cortex layers – primarily based on mouse studies – where the progenitors of each of the layers reside juxtaposed to the dermal papilla and where the daughter cells are differentiating and feeding the different layers^{13,14,18–21,48} (Fig. 1c, and Discussion).

Mapping single cell volume and dynamics in the follicle

To quantify the volume increase of individual cells after mitosis and during their upwards movement, we acquired 3D third-harmonic generation (THG) images on different follicles allowing the detection of the cell-walls of individual cells (Fig. 4a, green). This allowed the reconstruction of individual 3D cell membrane shapes and thus their volume along the IRS streamlines (Fig. 4a). Using this method, we observe that cells contributing to the IRS layer increase size as they migrate upwards. We validate this observation with an alternative way of measuring the cell volume using Hoechst-33258 as nuclear marker to determine the inter-cell nuclei distances (Supplementary Fig. 1e–f). Using this method, we measure a similar volume increase of upwards migrating cells of the cortex and the IRS during the first 150 μm above the dermal papilla (Fig. 4b). After that, the volume increase reaches a plateau, suggesting that once the cells become differentiated, their volume remains constant. This aligns with observations of endoreplication-associated volume increase in cortex cells⁵⁰, however, further investigation would be needed to confirm the presence of endoreplication in our human hair follicle culture model.

The velocity fields in the bulb should form a continuum with the upward moving cell speeds observed in the supra-bulbar area (Fig. 2a). To get a global picture of the velocity fields, we imaged the entire follicle segment including bulbar and supra-bulbar regions (Fig. 4c). The corresponding PIV analysis shows that the observed high speeds of IRS feeding cells in the bulb remain relatively high at the location of the IRS in the supra-bulbar region (Fig. 4c, right panel, See Supplementary Fig. 4h for additional PIVs). To quantify the speed progression along the hair follicle axis, we manually track individual cells at different heights above the tip of the dermal papilla and at different radial distances of the follicle (Fig. 4d, Supplementary Movie 9). Consistent with the PIV analysis, and consistent with the overall analysis in Supplementary Fig. 3l–n the averaged velocity of IRS cells shows higher speeds when compared to the cortex cells, especially for positions up to 100 μm above the dermal papilla (Fig. 4e). At higher positions, the cell migration speed starts to be equivalent between the different layers (Fig. 4e, grey line).

Mathematical modelling of follicular cell dynamics

The above quantified cell flows are reminiscent of a fluid flow within a cylindrical geometry with follicular shape. Before setting up a numerical simulation, we checked the data for self-consistency verifying that the continuity-equation holds. This equation states that the rate of volume generated within the system via mitosis and growing cells equals the balance between fluxes leaving the system and fluxes entering the system. This can be written in mathematical terms as:

$$\frac{d(\rho V)}{dt} = - \int_A \rho \vec{v} d\vec{A} \quad (1)$$

The left side of the equation considers all mass changes occurring inside a closed area A surrounding the follicle, e.g. in our case all mass changes via mitotic events and growing cells (Supplementary Fig. 4a). The right side of the equation considers all inwards and outwards

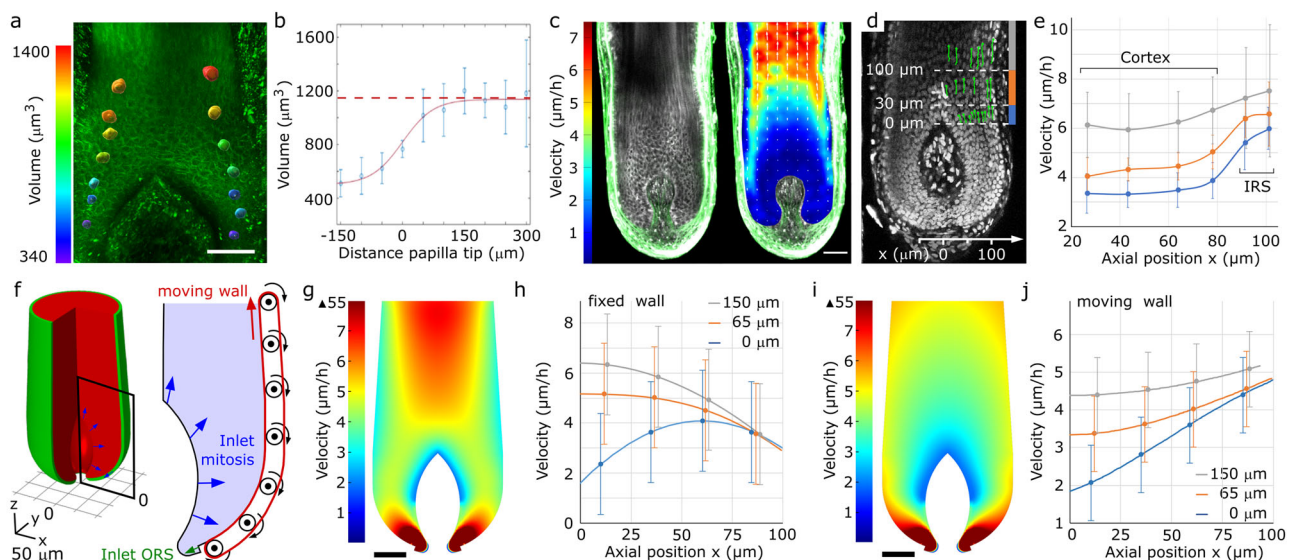


Fig. 4 | Map of cell dynamics in the entire follicle and numerical simulation.

a Longitudinal image section of follicular membrane architecture determined via the THG signal (green). The color-code corresponds to the volume of reconstructed, individual cells. **b** Estimated cell volume based on the nucleus-nucleus distances, data points denote mean $\pm$ SD, $N = 10$ follicles. **c** Cellular streamlines of entire hair follicle, revealed by the temporal projection of 10 time points. The corresponding PIV is shown on the right. **d** Tracking of individual cells along the axial position (x) of the follicle for three different zones at increasing height above the dermal papilla tip. **e** Average speed profiles for cells tracked at a relative height between 0 and 30 μm is shown in blue, between 30 and 100 μm in orange and

above 100 μm in grey (data points denote mean $\pm$ 95% confidence interval of the SEM, $N = 5$ follicles). **f** 3D view of the axisymmetric follicle computational model. The right panel shows a 2D-detail of the inner domain (cortex + IRS) close to the dermal papilla, where the inlet velocities accounting for mitosis and IRS, as well as a moving wall are emphasized. **g** Calculated speed profiles for a fixed wall ($v = 0 \mu\text{m/h}$). **h** Corresponding velocity profiles at three different heights of the follicle. The error bars are derived from the experimental uncertainty linked to the used inlet velocities given in $\pm$ SD. **i** Calculated speed profiles for an upward moving wall ($v = 6.75 \mu\text{m/h}$) and **j** corresponding velocity profiles with error bars as defined in **h**. Scale bars, 50 μm . Source data are provided as a Source Data file.

flowing cells crossing the area A, with the downwards moving ORS cells denoting the incoming flux, and the upwards moving cortex and IRS layers corresponding to the outward flux. Based on the experimentally measured parameters, we calculated for both sides: 63 ± 9 cells/h = 65 ± 30 cells/h (see Supplementary Methods). Within the given standard error of the mean, the continuity equation thus holds true.

We next set-up a 2D-axisymmetric computational model based on the Navier-Stokes equation, where the collective cell motion is described as an incompressible viscous fluid. Such continuum-models have already been used to describe limb bud formation⁴ and are well adapted to the human hair follicle with a rigid outer wall given by the connective tissue. We designed an average follicle shape with different inlets due to mitotic events and due to the downwards ORS migration based on our experimental data (Fig. 4f). We also included a movable wall at the outer border of the IRS layer for which we can fix the direction and speed (drawn as red ORS conveyor belt in Fig. 4f). Considering this wall as fixed with a speed $v = 0$ $\mu\text{m}/\text{h}$, we obtain speed profiles that do not match the experimental data (Fig. 4g, h, see Methods and Supplementary Fig. 4i–l for additional conditions). However, assuming a wall moving upwards at a speed of $v = 6.75$ $\mu\text{m}/\text{h}$ leads to velocity fields that approximate the experimental data (Fig. 4i–j). This assumption of an upwards moving wall can be physically interpreted as an active force superimposed to the classical fluid-like passive forces (pression and friction), which is located at the border between downward moving ORS cells and the upward moving layers.

A pulling mechanism implicated in hair growth

To check whether a pulling force acts at the interface between ORS and upward moving layers or if these are actively pushed by mitosis in the matrix, we first experimentally deleted possible factors leading to hair growth. To suppress a possible mitotic pressure arising from dividing cells in the matrix we surgically removed the bulbar region of the hair follicle after an initial *ex vivo* growth phase (Fig. 5a, 2. cut). Over the following 24 h we observed a continued upward movement of IRS and cortex layers, indicating an active pulling mechanism located above the papillary tip (Fig. 5b, Supplementary Movie 10). To determine the radial layer position responsible for this active sliding, we immunolabeled different follicles after the experiment with anti-Keratin14 antibody to stain the ORS layer (green) and anti-Keratin75 to stain the companion layer (magenta, Fig. 5b). Parts of the ORS layer remained connected to the static connective tissue, while the companion layer moved upwards together with parts of the ORS, IRS and cortex, indicating that the sliding occurs within the ORS itself, moving relatively to the companion layer and to the connected tissue sheath.

One possible mechanism for this pulling force could be the migration of the ORS cells themselves. The ORS layer is known to be rich in actin, organized as an intercellular network^{21,52}. Actin has an important role in maintaining the mechanical integrity of the cellular and tissular structures as well as in cell migration. Latrunculin B, an actin depolymerizing agent and an inhibitor for actin polymerization, is known to reduce cell contractile forces without completely depolymerizing the actin fibers at a concentration of 500 nM⁵³. The impact of 500 nM of Latrunculin B on bulb-less hair follicles is quantified in

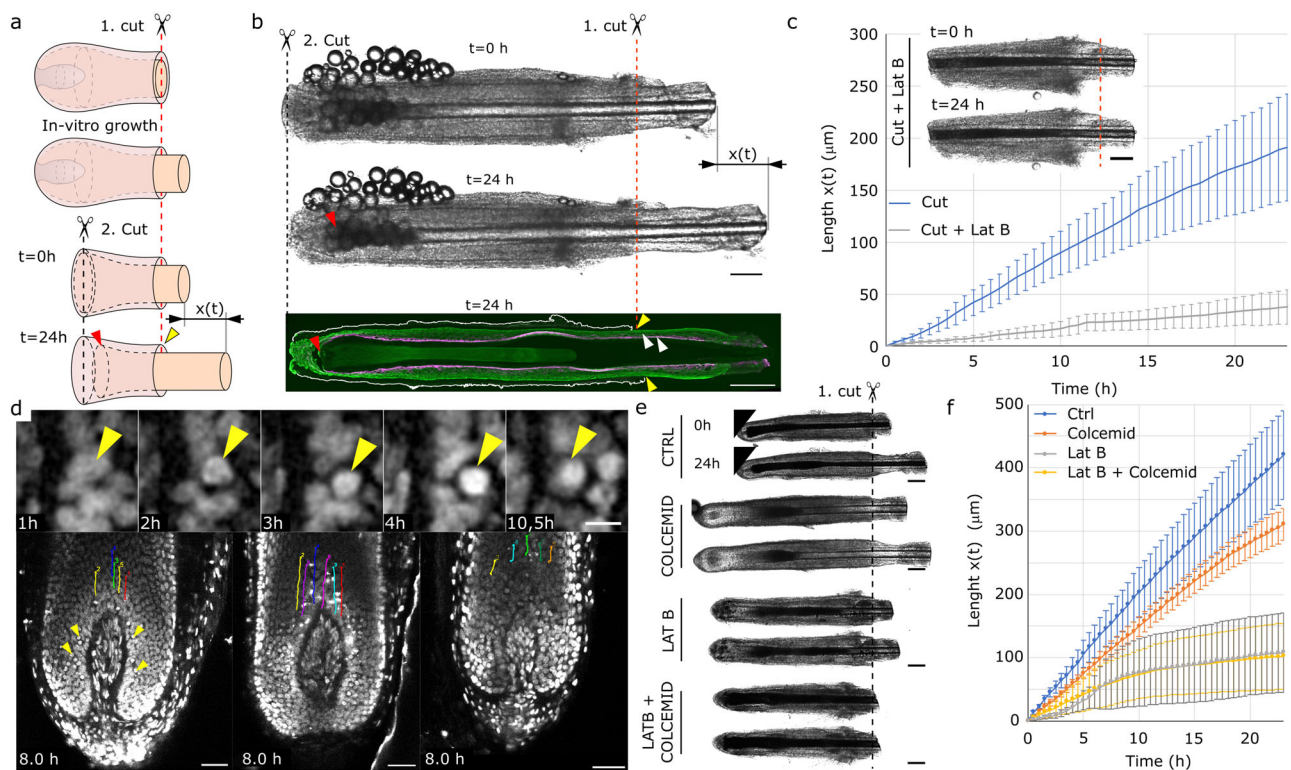


Fig. 5 | Experimental depletion of different hair growth drivers. a Experimental protocol eliminating the proliferating matrix via mechanical cutting.

b Representative brightfield image of a follicle directly after cutting off the bulbar area (2. cut, $t = 0$ h) and 24 h later, highlighting the continuous growth $x(t)$. Staining of a similar follicle with K14 (ORS, green) and K75 (companion layer, magenta), the white line traces the connective tissue ($N = 5$). The yellow arrowhead denotes position of 1st cut, the red arrowhead shows position of 2nd cut on the cortex which moved inwards, and the white arrows denote the extrusion of the internal layers including the companion layer in magenta. **c** Average length increased over time after the 2nd cut (blue, $N = 8$) and if Latrunculin B is added additionally (grey, $N = 9$).

Data points denote mean $\pm$ 95% confidence interval of the SEM. The inset shows a typical brightfield image for a Latrunculin B treated follicle. **d** Intact follicle (without 2nd cut) when treated with colcemid. Upper row shows zoom on matrix cell unable to divide (yellow arrow), lower row shows manual tracks (colored) on upward moving cells when mitosis is blocked (yellow arrows, $N = 4$). **e** Brightfield images and **f** average length increase for dissected follicles without 2nd cut either without treatment (ctrl, $N = 5$) or with addition of colcemid ($N = 6$), Latrunculin B ($N = 5$) or both ($N = 8$). Data points denote mean $\pm$ 95% confidence interval of the SEM. Scale bars, 200 μm (**b**, **c**), 50 μm (**d**). Source data are provided as a Source Data file.

Fig. 5c. We measured the fiber growth $x(t)$ over time for untreated, bulb-less follicles (blue) and when Latrunculin B is added (grey) which reduces the average growth speed by 80% (Fig. 5d).

To test the role of the mitoses on the upwards growing layers, we kept the dissected follicle intact and suppressed the mitotic activity of the cells using demecolcine (colcemid) at 0.25 $\mu\text{g}/\text{mL}$. As expected, we can observe that at this concentration the cells are unable to divide for several hours (Fig. 5d, upper row, Supplementary Movie 11); we could not observe the formation of a mitotic spindle in any cell ($N=4$). However, the growing layers of the follicle continued to flow upwards, as shown by the tracks of individual cells (Fig. 5d, lower row). This continuous movement together with the blocked mitosis leads to a decrease in cell density in the matrix (Supplementary Fig. 5, Supplementary Movie 11). Using brightfield microscopy on entire follicles with colcemid-inhibited mitosis, we observe that their growth speed drops only slightly ($\sim 24\%$, Fig. 5e, f). In comparison, when Latrunculin-B is applied alone or together with colcemid, the growth speed drops drastically ($\sim 82\%$).

Taken together, these results demonstrate that the mitoses of matrix cells are not the main force driving the upward migration of the cortex, IRS and companion layers. Rather, we propose an actin-mediated pulling force located in the ORS layer as a major element driving hair growth.

Discussion

The hair follicle offers a unique model for studying orchestrated cellular behaviors. One limitation is the challenge of tracking individual hair follicle cells over time. In this study, we map the primary human hair follicle cell fluxes, showing that IRS cells migrate upwards faster than cortex cells, which we attribute to forces applied by the ORS that pull the companion layer, along with the IRS and cortex, upwards. This pattern of faster IRS upward migration has been observed in murine¹³, sheep⁵⁴ and human follicles⁴⁹ in-vivo, further supporting our model of an active pulling force likely induced by coordinated downwards ORS migration.

These movements should be reflected by underlying cell-cell interactions and structural coordination: Within the hair follicle, relative motion between the ORS and companion layer is mediated by desmosomal junctions connecting the IRS, Henle and Companion layers^{55,56}, along with specialised structures called *Flügelzellen*⁵⁷, that may stabilize the alignment of the IRS and companion layer, facilitating their coordinated upward movement and ensuring the structural integrity of the follicle's inner layers during this directed movement. The companion layer-ORS interface has markedly fewer adhesion complexes and keratin filaments^{12,55} allowing for relative motion. The dynamics observed in our study align with observations indicating that the Companion layer shares a growth pattern similar to the IRS and is part of the inner layers¹⁴. As differentiation progresses, adhesive binding between these layers becomes tighter, regulated by desmoglein 4-dependent attachment structures, resulting in a more unified structure^{55,57–59}.

Consistent with their migration trajectory, IRS and companion cells move upwards with the growing hair but ultimately will not contribute to the hair shaft itself. Instead, these cells undergo terminal differentiation and are shed in the upper follicle regions, allowing the hair shaft (medulla, cortex, and cuticle) to continue its unobstructed growth⁶⁰.

As previously observed^{51,52}, actin filaments are present in the ORS and show an inter-cellular circumferential arrangement perpendicular to the growth axis in the supra-bulbar region, whereas they become rectilinear on the bulbar region. This structural arrangement correlates with the distinct cell migration directions we observed in these regions, analogous to the epithelial rotation in *Drosophila* egg chamber elongation, where collective cell motion perpendicular to the elongation axis facilitates tissue elongation⁶¹. The coexistence of

clockwise and counterclockwise rotations may be influenced by mechanisms similar to nematic-like interactions⁶², where cell shape, cell-cell, and cell-ECM interactions likely coordinate these dynamics.

During catagen, the follicle regresses and the shaft together with follicular cells moves upwards towards the epithelium, while the dermal papilla remains attached to the vascular stalk. Different theories on the underlying forces have been proposed that can be divided in “push” and “pull” models¹⁰. The push-models postulate a pushing force either due to mitotic pressure or via a “squeezing” mechanism due to a reduction of the radius of the connective tissue sheath. The pull models postulate an upwards force due to either ORS migration (immigration theory) or via occurring apoptosis (apoptotic force). Our findings suggest that during anagen, an actin-dependent mechanism within the ORS, exerts an upward pulling force on the companion layer and other inner layers. This force could be due to the ORS downward migration itself² e.g., implicating cell internal, actin based retrograde flow at the ORS-Companion layer interface, which could also play a role during the catagen regression phase. As already suggested earlier, such a shear force during anagen would also explain the upwards tilt of hair cuticle cells⁴⁹. Although the Latrunculin B treatments (Fig. 5) support this model, the treatment cannot target specifically ORS cells, but rather affects all hair follicle cells and further research is needed to pinpoint the exact actin-driven process contributing to hair fibre extrusion. However, our data, together with the faster upwards movement of the IRS layer relative to the cortex previously documented in vivo in humans via tritiated thymidine assays⁴⁹, indicate a physiological, active pulling force located in the outer layers during anagen.

In rodents, differentiation patterns of cells in the bulbar region have already been mapped: Previous studies have characterized the homeostasis of the mouse hair follicle inner layers during anagen, with progenitors juxtaposed to the dermal papilla that are spatially primed to contribute to a specific inner layer^{13–15,18} (Fig. 1c). Our observations of cell fluxes in the bulb area (Fig. 3e–g, Supplementary Fig. 4b–d) support a similar mechanism of stereotyped organization of inner layers' progenitors in human hair follicles.

The rate and orientation of mitosis are known to be influenced by local force fields, with mitosis playing a role in reducing internal tissue stress⁶³. We observe that the companion layer and IRS progenitors exhibit higher mitotic rates than the cuticle and medulla progenitors (Fig. 3d), which could be attributed to these force fields. Additionally, signals from the dermal papilla and the connective tissue in contact with ORS cells at the proximal end of the bulb, as well as adipose tissue signaling, likely influence these orchestrated mitotic events^{64,65}.

We observe ORS cells continuously moving downward into the bulb and arriving at the lower proximal cup region (Fig. 2c, white arrows, Supplementary Movie 12). Supported by our cell tracking results, our fluid dynamics description indicates that these cells incorporate the later upwards-moving inner layers (see Supplementary Methods). While we cannot track a single ORS cell for extended periods in this region, we observe upwards moving cells in the proximal cup region (Fig. 2c yellow and green arrow, Supplementary Movie 12) with some cells detaching from the dermal papilla surface (yellow track in Fig. 2c, Supplementary Movie 13) and others not (green track in Fig. 2c, Supplementary Movie 14). PIV analysis around this lower proximal cup region (See Fig. 3g, Supplementary Fig. 4b–d) show flow profiles indicating that a subset of ORS cells moves deeper into the hair bulb, while another subset switches direction and migrates directly outwards of the lower bulb region later forming the outer layers (Supplementary Fig. 4, Movies 12–14). A previous study using intravital imaging of anagen mouse hair follicles in live mice has shown that in normal homeostasis, the ORS cells move into the matrix through the lower proximal cup and fuel the germinative layer to produce the differentiated inner lineages¹⁸. Our study suggests that ORS cells enter the lower bulb region in humans but longer tracking times together with 3D tracking are needed to quantify the exact relative fates of ORS

cells entering the bulb. Based on our current data, we can give estimates for the relative amounts of ORS cells contributing to the germinative layer as previously observed in murine hair follicles¹⁸, as well as for the amount of ORS cells that directly differentiate into the companion layer and move upwards, or that undergo cell death (See Supplementary Methods). These findings provide insights into the potential contribution of the outer layer cells to the inner layers, and refine our understanding of lineage plasticity in the hair follicle suggesting these mechanisms are conserved between human as well as mouse hair follicles¹⁸ and mouse vibrissae⁴⁵.

The combination of ex vivo 3D+t imaging and mathematical modelling has provided a high-resolution mapping of cellular behaviour within the human hair follicle, revealing critical insights into the roles of ORS dynamics and coordinated mitoses in the upward migration of inner hair follicle layers. Ex vivo organ culture is currently the only human model-system allowing for this high-resolution mapping (See Supplementary Discussion for a review on the physiological relevance, advantages and limitations of this model). However, its application in live imaging also presents some specific limitations, which may be overcome through future technological advancements. A key limitation is the short culture duration due to phototoxicity (~20 h) before signs of an anagen-to-pseudo-catagen transition emerge (see Supplementary Methods), restricting the tracking length during which single cells can be followed during anagen and allowing only preliminary estimates of ORS fluxes in the lower proximal cup region. Future imaging with improved signal-to-noise ratio, reduced phototoxicity and thus longer movie times, as well as 3D-PIV analysis⁶⁶ may better resolve these dynamics. Additionally, targeted labelling of specific cells ex vivo would enhance the tracking of early cell differentiation and would allow to study other, smaller cell populations such as immune cells, that might be present in the follicle bulb (See Supplementary Movie 15 for examples of different cellular sub-populations with distinct dynamics). The current approach represents the most advanced model available to date, enabling us to uncover key cellular movements; future technological developments will further enhance resolution and allow deeper insights into the dynamic organization of the hair follicle.

Methods

Human hair follicle microdissection and culture

Normal human scalp skin was collected from surgical residues from adult healthy volunteers in France undergoing reconstructive or aesthetics surgery, with the patients' written informed consent in accordance with the Helsinki Declaration and with Article L. 1243-4 of the French Public Health Code. Tissue collection is performed by Alphenyx (Marseille, France) with authorizations from the French Ministry of Research under the license number AC-2014-2141 and its latest renewal AC-2024-6570, including a validation of the Bioethics Unit of the French Ministry of Research. Exportation of derivative models from Human Biological Samples for Research Use Only is performed under authorizations from the French Ministry of Research under the license number IE-2014-749 and its latest renewal IE-2024-4099. Samples were anonymized before their reception by the authors and in compliance with the recommendations of the CNIL and with the French and European GDPR regulations, only age, sex assigned at birth, phototypes, skin types and anatomical site of samples were specified to the authors.

Human scalp hair follicles were isolated under a stereomicroscope with scalpel and fine forceps, as previously described²². Hair follicles were then cultured in William's E medium (Gibco, Paisley, UK) supplemented with 2 mM L-glutamine (Gibco, Paisley, UK), 10 mg/mL insulin (Sigma-Aldrich, Saint Louis, USA), 40 ng/mL hydrocortisone (Sigma-Aldrich, Saint Louis, USA) and 1% antibiotics (Gibco, Paisley, UK) and incubated at 37 °C under 5% CO₂, 95% of humidity. For multiphoton microscopy experiments, only non-pigmented hair follicles

were kept for imaging, avoiding absorption of excitation light and fluorescent emission light via melanin, to avoid excessive phototoxicity⁶⁷ as well as to avoid saturated fluorescent signal from melanin. For brightfield microscopy experiments, all kinds of pigmented follicles are used.

Human hair follicle holder for microscopy

Two types of sample holders were used to film the hair follicle. In the first one, mostly used for multiphoton microscopy experiments, hair follicles were placed either longitudinally in a groove (length 5 mm, width 200 µm, depth 100 µm) or vertically in a well (diameter 300 µm) on a custom-built sample holder, 3D-printed in a transparent photopolymer resin. This sample holder was filled with culture medium (supplemented William's) and sealed with vacuum grease with a 0.17 µm-thick coverslip. This allows the hair follicles to grow for more than 64 h with brightfield microscopy minimizing phototoxicity, and over 22 h with multiphoton microscopy. The second one has been used mainly for brightfield microscopy experiments: A silicone isolator forming a well (Grace Bio-Labs) is fixed on a glass slide (Menzel-Gläser Superfrost), filled with supplemented William's medium and a maximum of 5 hair follicles, and cover with a 0.17 µm-thick coverslip (Menzel-Gläser #1).

Live imaging

A droplet of immersion oil (refractive index of 1.33) is placed on the coverslip of one of the sealed sample holders containing from one to five human hair follicles. Multiphoton fluorescence imaging was carried out on a Leica SP8 multiphoton microscope with two non-descanned Hybrid detectors (filters LP405 and BP 380/14), coupled with a Spectra Physics Insight X3 femtosecond laser set at 760 nm, and a 25x objective (HC FLUOTAR L W VISIR, 0.95NA, Leica). Five different sources of contrast were used to image the hair follicle: the autofluorescence of the cell's NADH, the emission of second harmonic generation (SHG) by the collagen, the emission of third harmonic generation (THG) mainly by cell membranes, the Hoechst signal from the labelled nuclei, and the transmission of brightfield illumination. To label nuclei, hair follicles were incubated for 30 to 45 min with Hoechst 33258 (life Technologies, stock at 10 mg/mL) diluted 2000x in supplemented William's medium. Then, the hair follicle was rinsed once in supplemented William's medium for 5 minutes. The hair follicles were then mounted in the 3D-printed sample holder and imaged under the multiphoton microscope.

3D+t images (z-step = 4 µm with multiphoton microscope to 30 µm with brightfield microscope) were acquired with time-steps ranging from 3 to 60 min, for up to 4 days. Time-lapse recordings were processed with Fiji⁶⁸. For each frame, the x,y and z drifts of the hair follicle were corrected to compensate for rotations around z-axis and translations in the three directions, with the plugin *Correct 3D Drift*⁶⁹.

Manual cell tracking and related statistics and reproducibility

We analyzed single cell dynamics only in follicles before the anagen – pseudo-catagen switch occurred (on average the first 22 h of the movies, see Supplementary Methods) to avoid artefacts of long-term live imaging experiments. In addition, if the anagen-pseudo-catagen switch is observed before 8 h of culture time the follicle is discarded. Following this procedure, we acquired a total of 22 follicles coming from 14 different donors in standard ex vivo culture conditions (without addition of drugs or other raw materials), usable for single cell-tracking (Supplementary Fig. 3). The globally observable production of the hair fiber must be related to the average movement of single cells making up the fiber. However, observations at the single cell level come with variability, due to inherent fluctuations in how individual cells behave. To describe “average” movements of single cells of the different layers (as in Fig. 2), we need to be sure that they are reproducibly observed. To this end, we quantified the variability of the

results using an analysis of variance approach (ANOVA) following a designed acquisition and analysis procedure outlined next:

To quantify ORS speeds perpendicular ($\text{ORS}^{\perp}$) to the hair fiber growth axis, we imaged longitudinally within the ORS layer of the hair follicle (Supplementary Fig. 3a, upper right) and determined single cell speeds via tracking in the supra-bulbar region (Supplementary Movie 4). To quantify the layer speeds of cortex, IRS and ORS parallel to the hair fiber growth axis ($\text{ORS}^{\parallel}$) we imaged in the middle of the follicle longitudinally (Supplementary Fig. 3a, bottom). Within these sections we defined the measurement areas related to the different layers based on the known hair follicle architecture and the layer differentiation obtained via live imaging as explained in Supplementary Fig. 1 (see Supplementary Fig. 3b for a schematic drawing). For each measurement zone we tracked between 3 and 12 (on average 5.5) single cells during a minimal time of 1.5 h and up to 14 h corresponding to a maximal track length of up to 100 μm using the *MtrackJ* plugin⁷⁰. Interestingly, we can observe a small sub-population of cells with dynamics that are distinct from the collective layer dynamics (see Supplementary Movie 15) which were excluded in our tracking analysis. Some of these cells might correspond to non-epithelial cells such as immune cells known to be present as a small sub-population in the proximal region^{71,72}. Following this design, we obtained $n = 371$ individual tracks measured from $N = 22$ different follicles coming from $D = 14$ different donors. Supplementary Fig. 3c shows the amount of acquired tracks within each zone of the single follicles. Note that while we try to acquire a complete 3D film of the entire follicle, in some cases not all layers can be captured, explaining some missing tracks for specific zones in Supplementary Fig. 3c.

The related measured tracking speeds are shown as boxplots (each point corresponds to one tracked cell) in Supplementary Fig. 3d–g, for cortex, IRS, $\text{ORS}^{\parallel}$, and $\text{ORS}^{\perp}$ movements respectively. The boxplots illustrate the measured variability of cell speeds within one single follicle, between single follicles as well as between different donors (if multiple follicles were obtained from one donor). We used ANOVA to perform a variance decomposition using a random effect model with a two-stage nested design⁷³. The estimated covariance parameters related to the variance within the follicle are 1,6 for cortex, 1,0 for IRS, 9,3 for $\text{ORS}^{\parallel}$, and 24 for $\text{ORS}^{\perp}$; the ones related to the variance between follicles are 1,7 for cortex, 0,3 for IRS, 2,9 for $\text{ORS}^{\parallel}$, and 4 for $\text{ORS}^{\perp}$; and the covariance parameters related to the variance between donors are 0,2 for cortex, 2,0 for IRS and 2,8 for $\text{ORS}^{\parallel}$. The values for each layer related to the variance within a follicle, between follicles and between donors are of the same order of magnitude, which is why we based our analysis on the mean from one single follicle independent of the donor. The measured cell tracks are summarized as boxplots in Fig. 2e where the mean velocities are $4.2 \pm 1.8 \mu\text{m}/\text{h}$ for cortex; $5.6 \pm 1.8 \mu\text{m}/\text{h}$ for the IRS; $-6.9 \pm 3.8 \mu\text{m}/\text{h}$ for $\text{ORS}^{\parallel}$ and $12.0 \pm 7.7 \mu\text{m}/\text{h}$ for $\text{ORS}^{\perp}$ all obtained from longitudinal imaging, with the error given as $\pm$ SD. (If not otherwise stated errors are given as $\pm$ SD in the text).

Tracking the $\text{ORS}^{\perp}$ speeds in a transversal setting (Supplementary Fig. 3a, upper left, Fig. 2a middle row) gives comparable speeds of $9.1 \pm 2.5 \mu\text{m}/\text{h}$, ($N = 3$). Tracking the $\text{ORS}^{\parallel}$ speed in the suprabulbar area (starting up to 200 micron above the dermal papilla tip, see e.g. Fig. 2a) gives an average speed of $-5.2 \pm 3.8 \mu\text{m}/\text{h}$, ($N = 5$).

PIV analysis gives comparable absolute velocity fields e.g. in the supra-bulbar region (compare e.g. Fig. 4e with Fig. 4c and Supplementary Fig. 4h). For example, we can estimate the upward speed of cortex and IRS cells measured along a line at a height of 60 microns above the papilla tip via PIV analysis to $3.5 \pm 2.6 \mu\text{m}/\text{h}$ and $4.9 \pm 3.4 \mu\text{m}/\text{h}$ respectively ($N = 6$), which is comparable to the values obtained via manual tracking given in Fig. 4e ($4.0 \pm 0.8 \mu\text{m}/\text{h}$ vs $6.5 \pm 1.3 \mu\text{m}/\text{h}$ and to the values obtained in Supplementary Fig. 3m ($3.8 \pm 1.7 \mu\text{m}/\text{h}$ vs $5.6 \pm 1.4 \mu\text{m}/\text{h}$). However, PIV analysis has some limitations of accuracy especially when data is acquired in granular biological media with

inherent intensity fluctuations^{74,75} which is why we based our calculations in the main text mainly on manual tracking results. The manual tracking method on the other hand might overestimate the speed values, since we focus on tracks that remain within a single longitudinal imaging plane, neglecting out of plane movements. The future application of 3D-PIV approaches⁶⁶ to analyze single cell dynamics during hair follicle growth might resolve these issues.

We can use the manual tracking data to check for possible effects when Hoechst staining was used as compared to measurements based on the autofluorescence signal. The related boxplots are shown in Supplementary Fig. 3h, with no significant difference detected between AF and Hoechst values measured in the cortex and $\text{ORS}^{\parallel}$ (t -test statistic: 1,2 with p -value 0.2 and $t = 0.2$ with $p = 0.8$, respectively) and a small difference ($5.2 \pm 1.7 \mu\text{m}/\text{h}$ vs $6.3 \pm 1.4 \mu\text{m}/\text{h}$, t -test statistic 2,7 with $p = 0.007$) between IRS speeds.

As can be seen in the boxplots in the Supplementary Fig. 3d–g, there is variability for each layer speed when comparing single follicles. We wanted to see if there is a correlation between the layer speeds within one single follicle, meaning if a follicle with a more dynamic cortex growth also shows higher speeds in the other layers. Supplementary Fig. 3i–k show pairwise correlations between different layer speeds with a Pearson correlation of -0.4 for Fig. 4i; -0.9 for Fig. 4j and 0.7 for Fig. 4k. This means that to reduce variability in our experimental design we can analyze relative differences within one follicle. Another way of reducing variability could be by constricting further the measurement area. To exemplify this, we wanted to compare the IRS cell speeds relative to the cortex cell speeds in a 100-micron high zone directly above the papilla tip as shown in Supplementary Fig. 3l. The absolute values of the measured speeds from this zone are shown in Supplementary Fig. 3m, where one point corresponds to the mean of at least 3 tracked cells of one follicle. The absolute IRS values in this region ($5.6 \pm 1.4 \mu\text{m}/\text{h}$) differ with a t -value of 3.3 (p -value 0.003) from the average cortex values in the region ($3.8 \pm 1.7 \mu\text{m}/\text{h}$). However, when looking at the differences within each follicle (grey lines in Supplementary Fig. 3m and Supplementary Fig. 3n) one observes in 16 out of 17 follicles higher IRS than cortex speeds and thus a significant average difference of $1.8 \pm 1.3 \mu\text{m}/\text{h}$ with a t -statistic of 5.4 and a p -value of 5×10^{-5} .

Particle imaging velocimetry (PIV) analysis of follicular layers

The cell movements of the different layers in the hair follicle were analyzed by particle imaging velocimetry (PIV), using PIVlab software on Matlab⁷⁶. It calculates the velocity vectors within hair follicle image pairs and can calculate multiple parameters of the flow pattern. Usually, a contrast-limited adaptive histogram equalization (CLAHE) is applied with a 20 pixels window's size on the different images of the movie, to increase the signal to noise ratio before the analysis. For few movies, a Wiener2 denoise filter has been applied with a window's size of 3 pixels. We used direct Fourier transform correlation algorithm. First, a 64×64 pixels interrogation area passes through all the image with a step of 16 pixels and calculates the displacement of the cells. Then, in a second pass, a smaller area with 32×32 pixels is used with a step of 16 pixels to get more vector resolution. The ORS and IRS layers of the hair follicle as well as the cortex and matrix were analyzed. In PIV analysis of the bulbar region (Fig. 3g) we were not able to detect the downward moving ORS cells due to the thin size of the ORS layer in the bulb and the rotational ORS cells movement taking them out-of-plane.

Mitosis analysis

Mitosis occurs typically over a time span of 1 hour. Following the Nyquist-theorem, we used an image acquisition frequency of 2 images/h ensuring that the main stages of mitotic events were covered. The mitoses were measured manually in 30% of the matrix volume centered around the equatorial plan, of four different follicles during 14.5 h of growth. We restricted the volume of matrix analyzed to be

able to observe all the mitoses occurring there, due to the better resolution of the microscope in the xy plane than in the z-axis. The orientation of the mitotic spindle and thus the direction of cell division can be detected using Hoechst-33258 nuclear staining (Fig. 3b). The Fiji plugins 3D Density⁷⁷ and 3D Viewer⁷⁸ were used to obtain a 3D representation of the density of mitosis in the hair follicle matrix during this period. All the angles of mitosis were measured with Fiji. To obtain average values, all the mitosis' measured on four different hair follicles were projected on the same half-follicle (Fig. 3d), considering the axisymmetry of the hair follicle.

Numerical modeling of cell flow

The experimentally observed cell flow shows complex behaviour and will depend on multiple biophysical parameters including signal driven cell migration, local cell mechanics, cell division, and others. There are today different computational methods available that allow to model tissue development. Those models can be roughly subdivided in two families: continuum models and cell-based models that have each distinct advantages and disadvantages^{3,79–81}. Continuum models describe the tissue as a continuous visco-elastic material. Related formalisms have the advantage of having only few, physically relevant parameters and being computationally more efficient on large simulation domains but the disadvantage that local cell-structures and their interactions cannot be implemented in a straightforward way. On long time-scales relative to the rate of mitosis and apoptosis, a hydrodynamic view of the tissue becomes applicable^{82,83} which can be modelled using the Navier-Stokes-equation. Given the very small Reynolds number for typical biological tissues (here $Re < 10^{-9}$, where we considered viscosities in the range $0.1\text{--}100\text{ Pa}\cdot\text{s}$ ^{84–86}) the inertial terms of the Navier-Stokes equation can be neglected⁸⁰, as done also for example for the theoretical description for limb bud formation^{4,87,88}. Also, such an approach was shown to be versatile when local variable viscosity terms and local growth sources are considered, that can be defined via morphogen diffusion⁸⁹. Compared to processes like limb and cyst formation or gastrulation, the dynamic changes occurring in an anagen follicle are distinct in the sense that we can consider the connective tissue sheet together with the papilla as a static structure surrounding the embedded, mobile cells. The assumption of hydrodynamic flow at least below $200\text{ }\mu\text{m}$ above the papilla tip, can be backed by the observation that those cells are not yet differentiated and are mechanically comparable to other living tissues^{38,90}.

We set-up a 2D-axisymmetric computational model implemented by the finite element commercial software Comsol Multiphysics. Such model is based on the Navier-Stokes equations where the collective cell motion is described as an incompressible viscous fluid. In this context, the velocity and pressure fields represent, respectively, the collective cell migration (passive transport) and the average contact forces, while the source terms simulate the cell mitosis. In a first step no additional active biochemical forces on the cells were modelled. The 2D-axisymmetric representation allowed us to strongly reduce the computational time while keeping a 3D description. It accounts for longitudinal and radial cell movements, while it neglects the azimuthal movements which are not the focus of the model. The model geometry (Fig. 4f, left) was developed starting from the measure of the shape of different static follicle sheaths and based on the average diameters and shapes given by the inner side of the connective tissue (Supplementary Fig. 1) as well as the shape of the dermal papilla. In practice, we modelled a papilla with height $160\text{ }\mu\text{m}$ and maximal diameter $140\text{ }\mu\text{m}$, a follicle having a height of $400\text{ }\mu\text{m}$ and a maximal diameter of $220\text{ }\mu\text{m}$. The computational model is composed by the ORS domain (in green in Fig. 4f), where the downwards cell velocity is treated as an input parameter and an inner domain (Cortex + IRS, in red). The interface between these two domains (Companion layer) is described as a wall that the cells cannot cross, except at the bottom, where the cells

composing the ORS enter the inner domain and are described as an inlet flux. To account for the abrupt inversion of speeds in this area we designed the wall to be movable with a fixable direction and speed (drawn as red conveyor belt around the ORS in Fig. 4f).

To account for the volume creation by mitosis as well as the cell volume increase during migration within the cortex while keeping the model description simple, we set an inlet velocity located at the border between the matrix and the papilla (Fig. 4f, right panel). The angle and rate of the incoming flux is based on the experimentally determined data (Fig. 3d). As quantified in Figs. 3c and 4a, the volume creation is not only localized on the surface of the dermal papilla but continues along the streamlines due to newly occurring mitosis and to the volume increase of individual cells. To account for this, we set the final supra-bulbar volume of a cell along a streamline already at the initial inlet position. This assumption will result in an overestimation of the cell velocity within the papillary zone, but it should have minor impact for the cells further away from the dermal papillary tip, where the volume creation becomes smaller. With these assumptions, the model flux enters at specific angles and with specific amplitude such that (1) the flux in the upper papilla is $19,800\text{ }\mu\text{m}^3/\text{h}$, namely 19 mitosis/h, according to Fig. 3d; (2) the flux in the lower papilla is $45,500\text{ }\mu\text{m}^3/\text{h}$, namely 44 mitosis/h; (3) the angles of the inlet velocity vary between 0° and 45° , matching the measured angle distribution of cell divisions given in Fig. 3d.

Several configurations have been explored, where the wall is considered either fixed or mobile, and where the companion layer cells can either (fully or partially) slide on the wall or not slide at all. The latter, called no-slip boundary condition, is a universally accepted assumption employed to describe the interface between a fluid and a solid and which is verified for most Newtonian systems but which can be violated in very specific configurations, such as a strongly confined gas flow or the motion of contact lines⁹¹. It is also an inaccurate assumption in the case of dense granular flows^{91–93}, where the grains may slip on the solid surface if the shear forces that they exert on it overcome the friction threshold. In our model, we thus assume that some slip between the inner cells and the wall (ORS) occurs (partial slip condition) with a slip length $L_p = 50\text{ }\mu\text{m}$, reflecting the discrete nature of the cell movement (behaving as a dense granular flow) behind the fluid-like Navier-Stokes description. The results corresponding to a fixed wall configuration using partial slip are reported in Fig. 4g,h and do not match the experimental observations. Note especially the strong discrepancy at the outer radial distance in Fig. 4h vs Fig. 4e. Considering the experimentally observed downward movement of the ORS cells we would need to set the wall-speed even to negative values in a passive system, which would further decrease the calculated velocity close to the wall and further increase the discrepancy between simulation and experiment. It is also important to note that an even stronger discrepancy from the experimental velocity field would be obtained for a no-slip boundary condition (i.e., strong friction, see Supplementary Fig. 4i and j for fixed wall). The perfect slip condition (no friction between the fixed wall and cells, Supplementary Fig. 4k, l) would be closer to experimental data but is unphysical.

The moving wall configuration shown in Fig. 4i,j (upwards speed of $6.75\text{ }\mu\text{m}/\text{h}$) with partial slip between cells and wall (slip length of $50\text{ }\mu\text{m}$) is the configuration which best matches the experimental data from Fig. 4e. In these conditions usually the slip length is a multiple of the particle (here the cell) diameter⁹². Indeed, in our simulations the slip length that best describes the experimental velocities is twice the cell diameter. Such boundary condition reflects the fact that the cells can partially slide on the wall with friction, according to the Coulomb's law. The walls upwards movement can be physically interpreted as an active biochemical force pulling the companion layer and the connected IRS cells upwards.

In the fluid description of the cellular movements using the Navier-Stokes equation, we never mention the viscosity. In fact, due to

the small Reynolds number and to the stationarity of the observed velocity field, the Navier-Stokes equation simplifies to the Stokes equation, which is characterized by the uniqueness of its solutions when the velocity field is known at the system's borders. This is, in fact, the case here where the cell velocity at the follicle's borders is measured and imposed as boundary condition of the computational model. This implies that there is only one possible velocity field that must be independent of the fluid's viscosity. We verified this numerically by comparing different simulations with different fluid viscosities.

Actin depolymerization and blockage of mitosis

To inhibit actin polymerization and disrupt its organization, Latrunculin-B (Sigma Aldrich) was used, at a final concentration of 500 nM⁵³ in supplemented William's medium (Gibco, Paisley, UK) during live imaging experiments using brightfield microscopy. To stop cell divisions, demecolcine (colcemid) solution (Sigma, stock solution at 10 µg/mL in HBBS) was used on dissected hair follicle at a final concentration of 0.25 µg/mL in supplemented William's medium, all along live imaging experiment. Demecolcine solution was used either with the sample holder for multiphoton microscopy, or in the second sample holder (see Human hair follicle holder for microscopy section) used for brightfield microscopy.

Follicular bulb cutting experiment

The dissected hair follicles were placed in a petri dish, in supplemented William's medium. The follicles were cut with a scalpel just above the dermal papilla. These bulb-less follicles were then placed on a glass slide with a silicone isolator forming a well filled with supplemented William's medium, and a glass coverslip. Then, they were filmed for more than 24 h under a brightfield microscope.

Immunostaining

Whole thickness human skin sections, intact whole hair follicles and cultured whole hair follicles cut from their bulb after the 24-h imaging experiment, were embedded in Tissue-Tek OCT compound (Sakura). The samples were frozen and sectioned longitudinally at 10 µm. All the sections of the sample are stored at −80 °C. To label these sections, we rehydrate them in Coplin type staining jar (Deltalab) filled with PBS for 15 min. The samples were fixed in ethanol at −20 °C for 10 min. The follicle sections were rinsed with PBS for 15 min and mounted on plastic coverplates (Eprelia Shandon) with 600 µL of PBS. Then, the samples were permeabilized for 5 min with 0.5% of Tween20 in PBS. To saturate the non-specific sites, the sections were incubated for 30 min with blocking buffer (10% goat serum and 0.05% Tween20 in PBS). The primary antibody was diluted in blocking buffer and incubated for 60 min at room temperature or overnight at 4 °C (Mouse IgG3 anti-K14 at 1/50, Chemicon Millipore; chicken anti-K14 at 1/5000, Covance/Biolegend; guinea pig anti-K75 at 1/500, Progen; guinea-pig anti-K31 at 1/200, Progen; mouse anti-Transglutaminase 1 at 1/2000, clone BC-1). The samples were rinsed 3 × 5 min in 0.05% Tween20 in PBS. The secondary antibody was diluted in blocking buffer for 45 min (anti-mouse IgG(H + L)-Alexa488 at 1/400, Molecular Probes; anti-mouse IgG(H + L)-Alexa555 at 1/300, Thermofisher; anti-chicken IgY-Alexa488 at 1/300, Thermofisher; anti-guinea pig-Cy3 at 1/200, Merck Millipore). All sections were counterstained with either WGA (1:50, W11261, Invitrogen) or DAPI (1:500, D1306, Invitrogen). The samples were rinsed 3 × 5 min in PBS with 0.05% of Tween20 and then mounted between slide and coverslip with Prolong Gold Antifade Reagent mounting liquid (Life Technologies).

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Source data for all images are provided as a Source Data file with this paper. Exemplary 2D live imaging raw data is available in the Supplementary Movies 1 to 17. Entire 3D live imaging raw data is available from the corresponding authors upon request. Source data are provided with this paper.

Code availability

For live-imaging analysis we used publicly available programs (either commercially available or free of use) as detailed in the Methods section. For numerical simulations we used the commercially available platform Scmsol Multiphysics.

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Author contributions

N.T., I.S. and T.B. conceived the study and wrote the manuscript. N.T. initiated project and performed all in vitro experiments except for IHC which was performed by G.G. and I.S.; Data was analyzed by N.T. and T.B.; R.S. performed mathematical modelling and simulation analyses. N.T., G.G., R.S., F.B., S.T., J.-F.M., I.S. and T.B. discussed data, protocols and research. N.T., I.S. and T.B. generated figures. T.B. and N.T. supervised the project and monitored the experiments. N.T., I.S. and T.B. wrote and revised the manuscript. All authors read and approved the final manuscript.

Competing interests

N.T., G.G., R.S., F.B., S.T., J.F.M. and T.B. are employees of L'Oréal Company. I.S. is a consultant for L'Oréal Advanced Research.

Additional information

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